

Name: _____

Lab partner: _____

Lab Section: _____

HEAT

Purpose Practice heat flow analysis Determine specific heat capacity of a material	Equipment calorimeter cup two glass beaker (250 mL or more) hot plate water ice cubes (at melting point) thermometer scale balance with weights paper towel
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[A] Preparation

In this experiment, you will start with warm water in an insulated metal cup. Cold water will be added, and the mixture will reach a thermal equilibrium. Before performing the experiment, answer some questions about this situation:

1. If you pour hot water in the metal cup and wait a few minutes, how should the temperature of the cup compare to that of the water? Why?

2. The warm water and the cup will *lose* heat, while the cold water *gains* heat. How does the heat lost by the warm water & cup compare to the heat gained by the cold water? Why?

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[B] Introduction

This experiment uses a calorimeter, which is specifically designed to minimize heat transfer with the external environment. The definition of 1.00 calorie is the amount of heat needed to change the temperature of 1.00 gram of water by 1.00° Celsius.

Before you begin, read the instructions of Part 1 carefully. The boxed part is time-sensitive: make sure to perform these steps quickly but accurately.

[C] Measurements and Analysis – Heat transfers between warm and cold water

C1. Data collection (Situation I)

1. Obtain approximately 75 mL of water in a beaker (“C”). Add some ice and stir occasionally.
 2. Meanwhile, fill a second beaker (“W”) with approximately 125 mL of water. Insert a thermometer and use a hot plate to heat the water to approximately 45°C.
 3. While the ice melts and the water warms up, use a scale balance to determine the mass of the empty, inner calorimeter cup. Record this in Table 1 below.
 4. When the water in beaker W is sufficiently warm, pour it in the calorimeter cup. Measure the mass of the cup with the water and record it in Table 1.
 5. Place the calorimeter cup in the calorimeter (for insulation).
 6. *Remove from beaker C any ice that has not yet melted.*
7. Measure and record the initial temperature of the warm water in the cup. Cover with a lid.
 8. Measure and record the initial temperature of the cold water in beaker C.
 9. Pour the water from beaker C into the calorimeter cup. Cover the cup. Insert the thermometer.
 10. Stir and wait until the temperature quits changing significantly.
 11. Record the final temperature of the water.
12. Measure and record the mass of the cup with the water.

From the measured masses, calculate the masses of cup, warm water, and cold water individually. Record them in Table 1 below.

Table 1. Masses and temperatures.

<i>measured mass</i>	(g)	<i>calculated mass</i>	(g)	<i>temperature</i>	(°C)
empty cup				initial temp. warm water	
cup + warm water		warm water		initial temp. cold water	
cup + warm water + cold water		cold water		final temp.	

C2. Analysis of Situation I

Recall that the amount of heat needed to change the temperature of 1.00 gram of water by 1.00° Celsius is, by definition, 1.00 calorie. In other words, $c_{\text{water}} = 1.00 \frac{\text{cal}}{\text{g}^\circ\text{C}}$.

1. Determine the amount of heat lost by the warm water in this experiment.

temperature change $\Delta T_{\text{ww}} =$

heat flow $Q_{\text{ww}} =$

2. Determine the amount of heat gained by the cold water in this experiment.

temperature change $\Delta T_{\text{cw}} =$

heat flow $Q_{\text{cw}} =$

3. Compare the two heat flows. Does your result match the expectation you wrote down on the first page of this lab?

4. You have likely observed a difference in the heat lost compared to the heat gained.
 - a. Identify at least two avenues by which heat is lost to the environment during this experiment.
 - b. Design a hypothetical experiment that would allow you to test your ideas from part (a). Describe it here.

Checkpoint: _____

C3. Data Collection for Situation II (different amounts of water)

Repeat the experiment from part I. This time, use about **125 mL** of cold water and **75 mL** of warm water. Use Table 2 and the template below to analyze the new data.

Table 2. Masses and temperatures.

<i>measured mass</i>	(g)	<i>calculated mass</i>	(g)	<i>temperature</i>	(°C)
empty cup				initial temp. warm water	
cup + warm water		warm water		initial temp. cold water	
cup + warm water + cold water		cold water		final temp.	

C4. Analysis of Situation II

warm water

temperature change $\Delta T_{\text{ww}} =$

heat flow $Q_{\text{ww}} =$

cold water

temperature change $\Delta T_{\text{cw}} =$

heat flow $Q_{\text{cw}} =$

comparison

[D] Measurements and Analysis - Heat Transfer between metal and water

D1. Data Collection (Situation III)

For this experiment, you need a 200-g metal object that has been sitting in melting ice for a while. That way, you know that the object will be at 0°C.

1. Fill a beaker ("W") with approx. 100 mL of water. Insert a thermometer and use a hot plate to heat the water to approximately 45°C.
2. When the water in the beaker is sufficiently warm, pour it in the calorimeter cup. Measure the mass of the cup with the water and record it in Table 3 below.
3. Place the calorimeter cup in the calorimeter (for insulation).
4. Take the 200-g object from the ice. Don't allow it to touch anything else.
5. Measure and record the initial temperature of the warm water in the cup.
6. Add the 200-g object to the cup. Cover with lid. Insert thermometer.
7. Stir and wait until the temperature stabilizes.
8. Record the final temperature of the water.

Complete Table 3. The mass of the empty cup you already know from Part I.

Table 3. Masses and temperatures.

<i>measured mass</i>	(g)	<i>calculated mass</i>	(g)	<i>temperature</i>	(°C)
cup				initial temp. warm water	
cup + warm water		warm water		initial temp. metal object	0.0°C
metal object	200 g			final temp.	

Type of metal (circle one): brass or stainless steel

D2. Analysis of Situation III

1. Determine the amount of heat lost by the warm water in this experiment.

temperature change $\Delta T_{\text{ww}} =$

heat flow $Q_{\text{ww}} =$

2. How much heat flowed into the metal object in this experiment? Explain how you determined this.

$Q_{\text{weight}} =$

By how much did the temperature of the metal object change?

temperature change $\Delta T_{\text{weight}} =$

3. Based on the previous answers, analyze the thermal properties of the material of the 200-g metal object by determining the specific heat of the metal. Show your work. Calculate the percent difference compared to the known values of $c_{\text{stainless}}=0.12 \text{ cal/g}^\circ\text{C}$ and $c_{\text{brass}}=0.092 \text{ cal/g}^\circ\text{C}$.

[E] Reflection

1. A group of students performs the same experiments you performed, but they stir the water very vigorously. How might this affect their final temperature? Why?
2. A 100.0 g sample of a substance is heated to 100.0°C and put in a calorimeter cup (having a negligible amount of heat absorption) containing 150.0 g of water at 25°C . The sample raises the temperature of the water to 32.1°C . Calculate the specific heat of the sample. Show all work.
(Hint: Recall your analysis of Situation III in this lab).